The Green Capital Gap

A deal-level analysis of cleantech venture funding in emerging economies, 2000–2012

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The Green Capital Gap

A deal-level analysis of cleantech venture funding in emerging economies, 2000–2012

Abstract

Emerging-market green entrepreneurship is almost absent from the open venture record, and what absence there is turns out not to be about green. Across 41,105 funding rounds from the Crunchbase 2013 Snapshot, classified by a three-tier green classifier with inter-tier agreement of $\kappa = 0.850$, an emerging-market company already indexed in the record is **half as likely to be green as an advanced-economy one**: 2.60% against 5.43%, a risk ratio of 0.479 [0.343, 0.669]. That gap survives every green definition, every bloc scheme, and a permutation test in which **zero of 1,000** label shuffles reach the observed value.

The gap is **not green-specific**. Across 23 sectors, emerging-market representation falls roughly in proportion to capital intensity, with an elasticity of -1.19 [-1.64, -0.75] explaining 0.60 of the cross-sector variance; biotech sits further below proportionate representation than cleantech does. Cleantech's residual from that fit is **positive** at $1.92\times$ [$0.75\times$, $4.91\times$], so a green-specific penalty is ruled out.

Three of the five hypotheses could not be resolved, each for a different and separately verifiable reason, and together with three further measurement failures they give **six independent routes to one conclusion: the capital-regime question is not answerable from open data.** That is the paper's finding, not its limitation.

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5. The double gap

5.1 Emerging markets are a fiftieth of the record and a fifth of world GDP

Over 2000–2012 emerging markets held 15.3–22.7% of world GDP and 1.7–6.4% of recorded venture deals, an under-representation of 2.8–8.8× (Figure 1). This is the coverage gap, and it is the reason the paper's headline result is measured *within* the dataset rather than against the world.

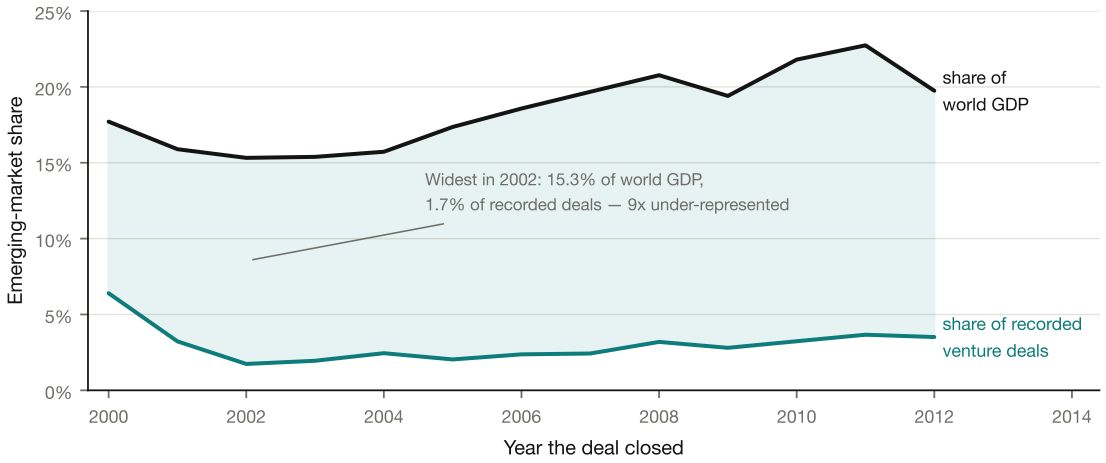
5.2 Conditional on being in the record, an emerging-market company is half as likely to be green

1,309 emerging-market deals carry 2.60% green [1.86, 3.61], against 5.43% [5.20, 5.66] across 37,896 advanced-economy deals. The difference is -2.83 percentage points [-3.72, -1.94]; the risk ratio is 0.479 [0.343, 0.669], z-test $p = 7.5\times10^{-6}$, Fisher's exact $p = 1.3\times10^{-6}$ (panel (a) of Figure 2).

Both numerator and denominator are drawn from companies Crunchbase chose to index, so the result does not depend on the coverage gap in §5.1. Counting every deal with no recorded

Emerging markets hold a fifth of world GDP and a fiftieth of the venture record

Emerging-market share of world GDP against its share of recorded venture deals, 2000–2012.



Source: Crunchbase 2013 Snapshot (CC BY 4.0); author's calculations. n = 41,105 deals. GDP shares are computed over the same country set the bloc panel covers, so the two series are comparable by construction. This is the denominator check for Figure 1: it establishes that the conditional green-rate gap is not this coverage gap restated.

Figure 1. Emerging-market share of world GDP against its share of recorded venture deals.

country as emerging-market — the most generous possible treatment — gives 3.46% and a risk ratio of 0.637, $p = 6.3\times10^{-6}$.

China, reported separately throughout, sits with the advanced economies rather than with emerging markets. Figure 3 shows the two rates by year, with every emerging-market cell annotated by its count.

Verdict: Supported. Falsified if the emerging-market green rate were indistinguishable from the advanced-economy rate, or if the gap vanished under the upper bound.

5.3 The gap is a capital-intensity gap

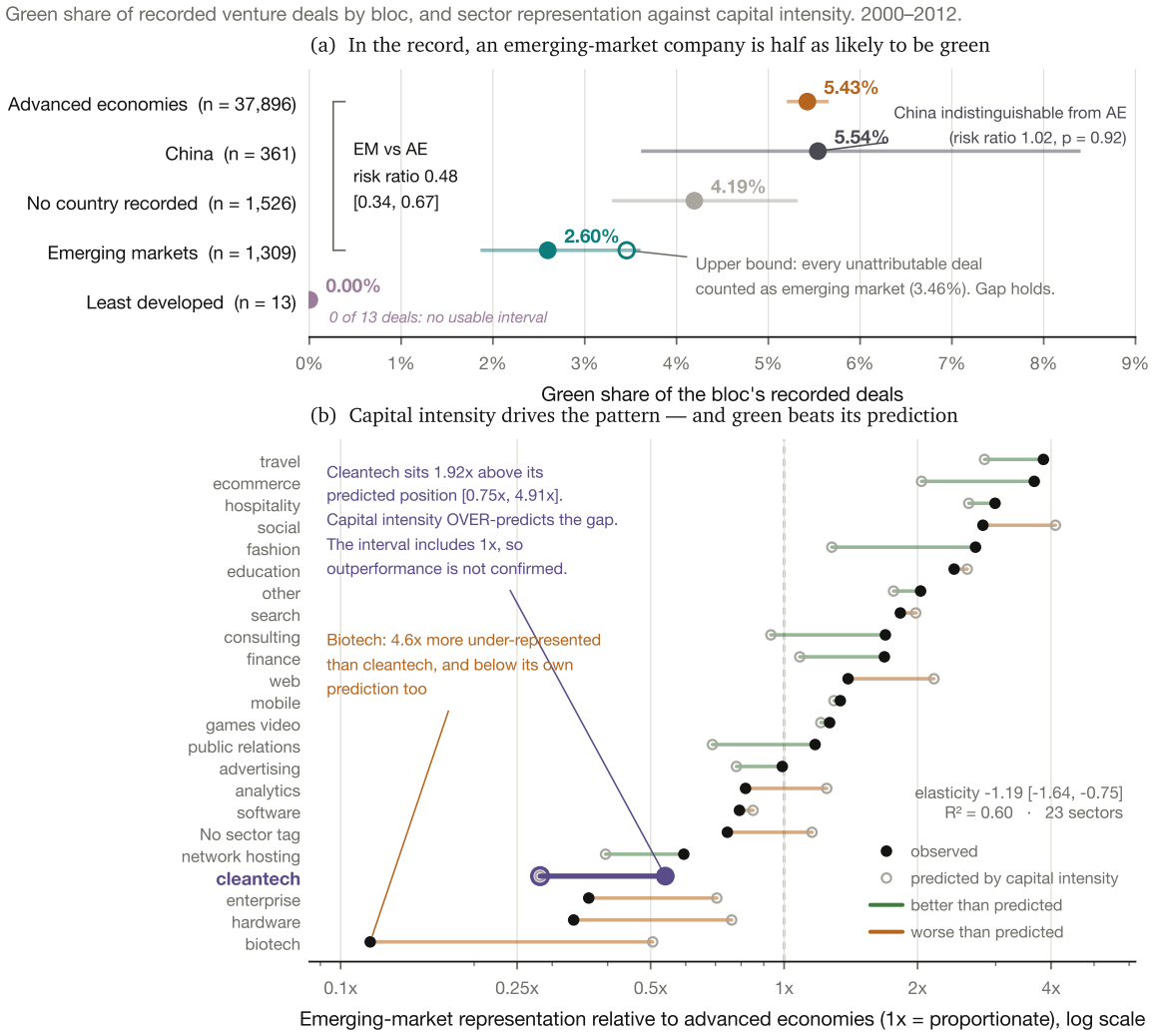
Emerging-market venture capital in this period concentrated in capital-light consumer internet and avoided capital-intensive sectors as a class. Regressing log emerging-market representation on log median advanced-economy deal size across 23 sectors gives an elasticity of -1.19 $[-1.64, -0.75]$, $p = 1.5\times10^{-5}$, $R^2 = 0.60$ (panel (b) of Figure 2). Doubling a sector's capital intensity roughly halves its emerging-market representation.

Cleantech sits **above** the position that fit predicts, at $1.92\times$ $[0.75\times, 4.91\times]$ of its predicted representation. The interval includes 1, so outperformance is not established; what is established is that there is no green-specific penalty once capital intensity is accounted for. Biotech, by contrast, sits below its own prediction.

Green also took a larger share of venture dollars than of venture deals from 2005 onward (Figure 4), which is consistent with green being capital-intensive without testing the cross-sector elasticity above.

5.4 The taxonomy cannot say what kind of green

The emerging-market green gap is real. It is not green-specific.



Source: Crunchbase 2013 Snapshot (CC BY 4.0); author's calculations. Panel (a) n = 41,105 deals, 2,174 green; bloc rows sum to the total. Panel (b) 23 sectors with ≥15 emerging-market and ≥30 advanced-economy deals; capital intensity is median advanced-economy deal size in constant 2015 USD, so the proxy cannot be contaminated by the emerging-market composition it explains. Intervals are Wilson score (a) and residual 95% CI (b). Headline is raw classifier output; bias-corrected risk ratio 0.44, curated 0.40 (Appendix C).

Figure 2. The double gap by bloc, and the sector representation ladder that qualifies it.

88.2% of emerging-market green deals carry no sub-domain more specific than “Clean Technology”, against 67.9% in advanced economies (Figure 5). The mechanism is direct: 26 of the 28 emerging-market green companies carry no other taxonomy tag at all. The planned sub-domain mix comparison was cut, because the measurement failure shown here makes it impossible.

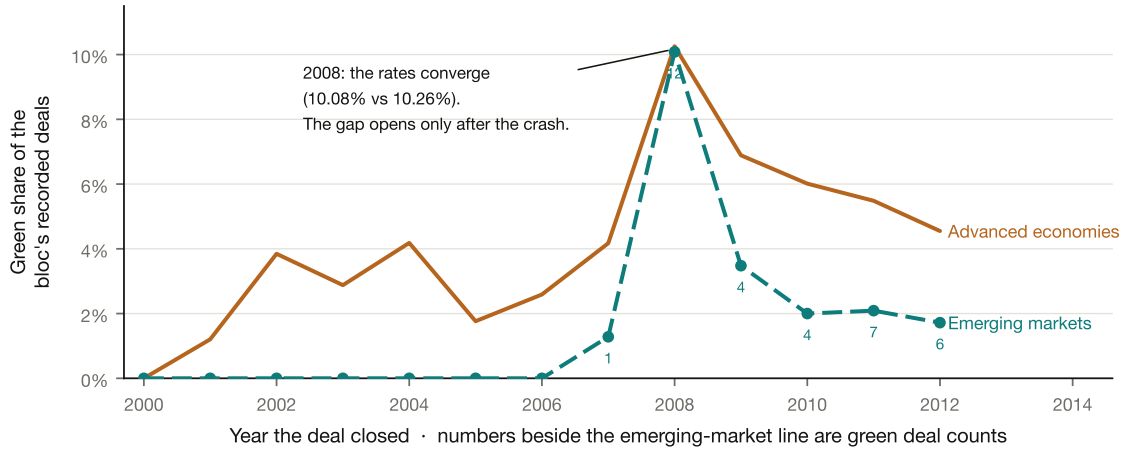
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6. The advanced-economy benchmark, and what each null says about the record

Advanced economies are the one bloc where the record is dense enough to test the structural hypotheses. They are the benchmark case for the capital-model mismatch Gaddy et al. identify. Each subsection closes by stating what its emerging-market arm shows about the record itself.

The two rates met in 2008, at the peak of Cleantech 1.0, then diverged

Green share of recorded deals by bloc and year, 2000–2012.



Source: Crunchbase 2013 Snapshot (CC BY 4.0); author's calculations. n = 41,105 deals. NO emerging-market year-cell reaches the minimum cell size of 30 green deals — the largest is 12, in 2008 — so no per-year test is reported and this figure is descriptive only. Counts are printed under each emerging-market point so the reader can see the precision available. Raw classifier output.

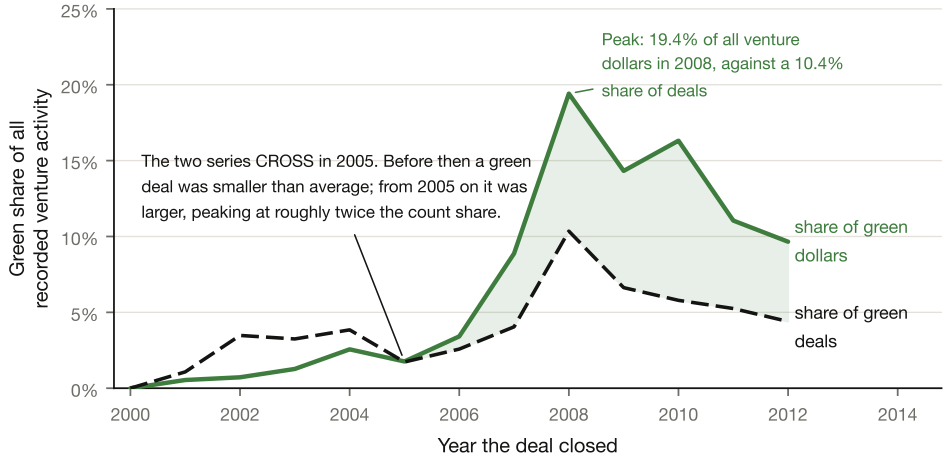
Figure 3. Green share of recorded deals by bloc and year, emerging-market counts annotated.

6.1 Green deal sizes fell far faster than non-green within the same stages

Between 2007 and 2012, within named funding stages, green deal sizes fell −50.4% against −1.1% for non-green — a difference of −49.3 percentage points [−69.3, −22.4], bootstrapped over 10,000 draws (Figure 6).

From 2005, green took a bigger share of venture dollars than of deals

Green as a share of all recorded deals and of all disclosed dollars, 2000–2012. Constant 2015 USD.

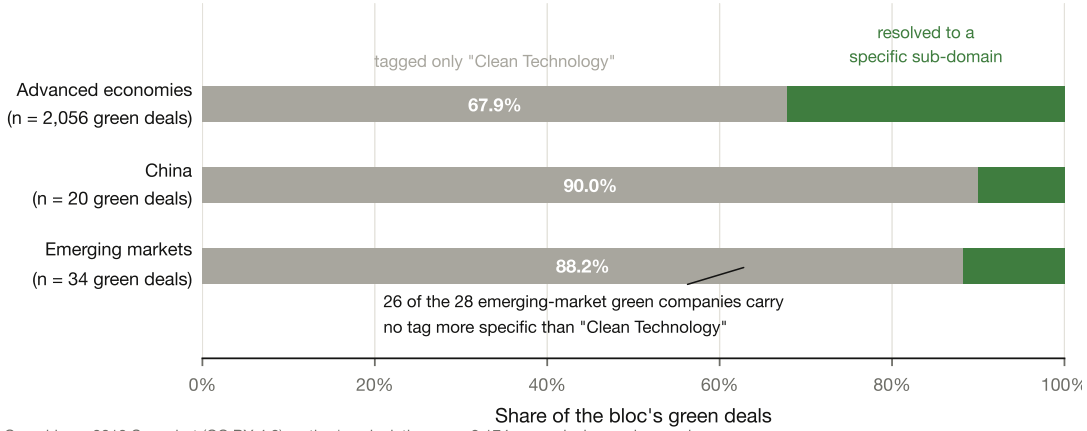


Source: Crunchbase 2013 Snapshot (CC BY 4.0); author's calculations. n = 41,105 deals, 2,174 green. The dollar series covers the disclosed subsample only, and both series are computed on that same subsample's denominator where dollars are concerned, so the comparison is internally consistent. A dollar share above the count share means the average green cheque exceeded the average cheque overall, which is consistent with green being capital-intensive; it does not test the cross-sector elasticity of Figure 1b. The ordering reverses before 2005, when green deals were SMALLER than average, on thin cells: 2001 n=1, 2002 n=4, 2003

Figure 4. Green as a share of all recorded deals and of all disclosed dollars.

The taxonomy cannot say what kind of green an emerging-market company is

Share of green deals resolved to a specific sub-domain, by bloc, 2000–2012.



Source: Crunchbase 2013 Snapshot (CC BY 4.0); author's calculations. n = 2,174 green deals carrying a sub-domain label. The eleven specific sub-domains are solar, wind/hydro/geothermal, biofuels, storage, efficiency and smart grid, electric mobility, water, waste and circular, sustainable agriculture, green building, and environmental monitoring. This figure replaces the planned emerging-market sub-domain mix comparison, which the measurement failure shown here makes impossible.

Figure 5. Share of green deals resolved to a specific sub-domain, by bloc.

The pooled median falls by about the same amount in both groups and settles nothing: it is composition-confounded for each, because the platform recorded far more small rounds as it matured. All 5 start-end pairs running to the close of the window are negative, spanning −69.9 to −21.5 percentage points, and the headline figure is their median, so the direction is robust to the choice of start year. It is **not** robust to the choice of end year: across all 17 admissible pairs the difference is negative in 71%, and every positive pair stops in 2006 or 2009–11, before the decline has run its course. The claim the grid supports is a direction that holds to the close of the window, with a magnitude that depends on where the comparison starts.

This is a size pattern, not an investor motive. The rotation out of hardware that Gaddy et al. document is cited here as external evidence, not as something this dataset shows.

6.2 The crisis hit green and spared non-green, but the test does not identify it

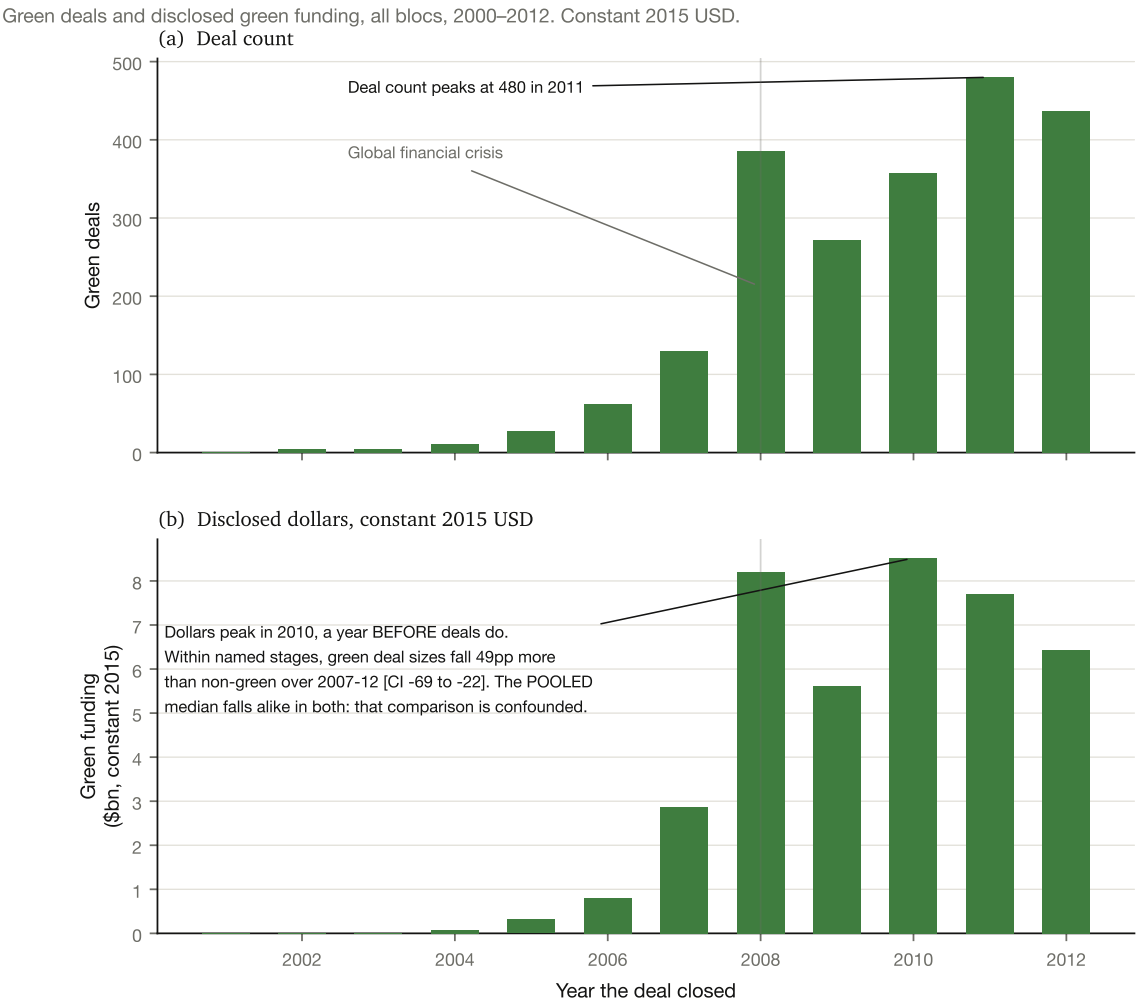
Across 2008–09, advanced-economy green deals contracted −26.5% while non-green **grew** 13.5% (Figure 7).

The difference-in-differences designed to test this does not identify it. Green deal volume was already growing 0.254 log-points per year faster than non-green before the event window, [0.101, 0.408], p = 0.001. Parallel trends fail, so the design never recovered a causal effect and there is no null to report.

Verdict: Not testable — identifying assumption failed. The contraction figures above are descriptive.

What the emerging-market arm shows about the record: it does not exist. The largest emerging-market green year-cell holds 12 deals.

Green deal sizes fell far faster than non-green within the same stages



Source: Crunchbase 2013 Snapshot (CC BY 4.0); author's calculations. n = 2,174 green deals, of which 1,994 carry a disclosed amount. Counts and dollars are reported as separate series because undisclosed-amount rates differ systematically by instrument and bloc; panel (b) covers only the disclosed subsample. Constant 2015 USD via the US GDP deflator. The cheque-size claim is the WITHIN-STAGE comparison: the pooled median falls about 72% for green and 73%

Figure 6. Green deals and disclosed green funding, in counts and constant 2015 USD.

6.3 Green capital is less dilutive than non-green

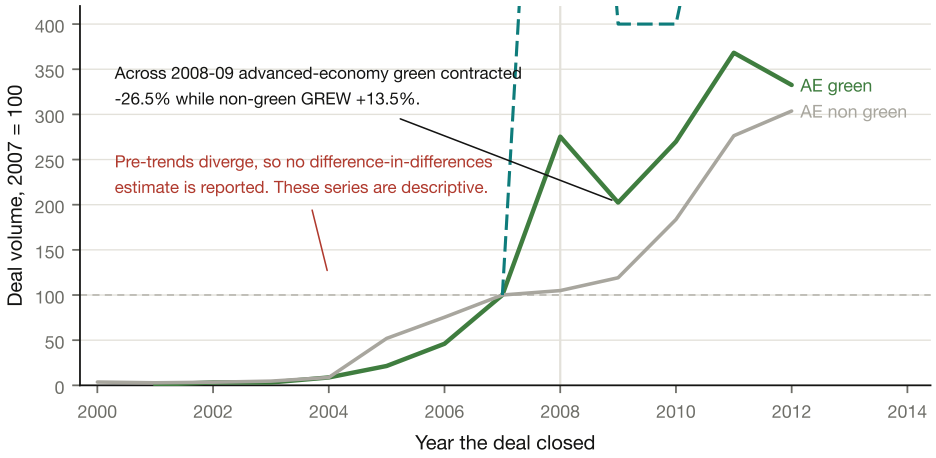
Within advanced economies, 10.41% of green deals use non-dilutive instruments — grants, debt, post-IPO debt — against 7.26% of non-green deals. The risk ratio is 1.433 [1.256, 1.636], $p = 1.2\times10^{-7}$ (Figure 8).

The χ^2 on the full instrument mix is also significant, but its Cramér's V of 0.027 falls below the conventional threshold for a small effect: with tens of thousands of deals the test detects a negligible association, and the risk ratio on the quantity the hypothesis is about is the interpretable figure.

Verdict: Supported for the advanced-economy benchmark case.

The crisis hit green venture funding and left non-green growing

Deal volume indexed to 2007, by bloc and green status, 2000–2012.



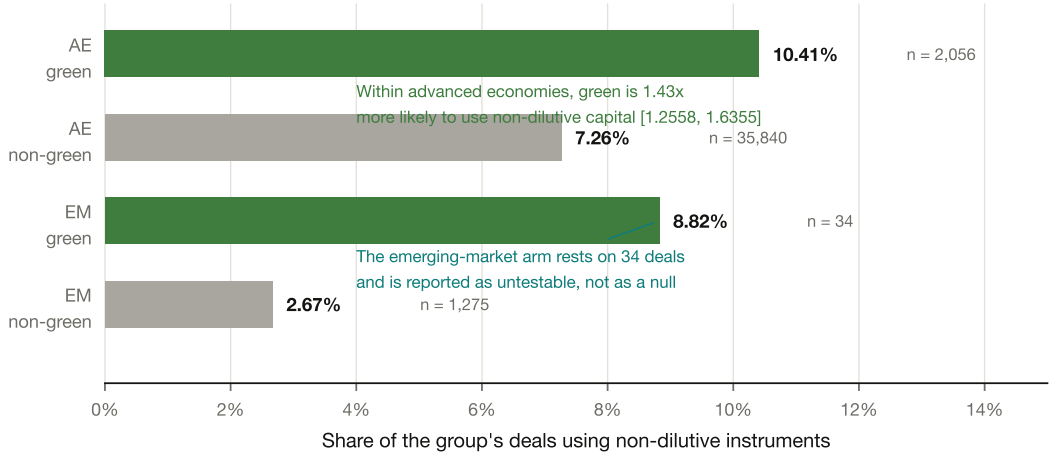
Source: Crunchbase 2013 Snapshot (CC BY 4.0); author's calculations. n = 41,105 deals. The emerging-market green series rests on 0–12 deals per year and is plotted for shape only. No difference-in-differences coefficient is shown: green volume was already growing +0.254 log-points per year faster than non-green before the event window (p = 0.0012), so parallel trends fail and the design does not identify a causal effect. Raw classifier output.

Figure 7. Deal volume indexed to 2007, by bloc and green status.

What the emerging-market arm shows about the record: it is null by construction. The record codes infrastructure project equity as ordinary venture rounds, which understates non-dilutive share on both sides and more so for emerging markets, where independent power producers are seven of 28 green companies. The direction of that bias was stated before the numbers were computed. On 34 deals the arm could not have settled the question either way.

Green ventures lean harder on grants and debt than non-green ones do

Non-dilutive share of deals by bloc and green status, 2000–2012.



Source: Crunchbase 2013 Snapshot (CC BY 4.0); author's calculations. Non-dilutive instruments are grants, debt financing and post-IPO debt. Advanced-economy comparison: risk ratio 1.433 [1.2558, 1.6355], p = 1.24e-07. Project finance that is debt or concessional in substance but coded as an equity round understates the non-dilutive share on both sides, and more so for emerging markets where independent power producers are seven of 28 green companies; that bias direction was stated before these numbers were computed. Raw classifier output.

Figure 8. Non-dilutive share of deals by bloc and green status.

6.4 The scale-up step is unobservable

Conditional on raising seed capital, advanced-economy green companies reached Series A with an odds ratio of 1.30 [0.65, 2.60] against non-green, p = 0.458. The interval spans 1 and identifies no direction.

It cannot. The advanced-economy green seed cohort holds 40 companies and the Series A to Series B pool — the specific missing-middle transition — holds 13. No model of that transition is reported, under any specification.

Verdict: Underpowered — cannot determine.

What the emerging-market arm shows about the record: there are 0 emerging-market green seed entrants across thirteen years. The scale-up step the missing-middle literature identifies as decisive cannot be observed at all.

6.5 Capital tracks market depth; whether it tracks need cannot be determined

Across 22 countries with enough green activity to measure, green funding per capita correlates with GDP per capita at ρ = 0.674 [0.298, 0.866] (Figure 9).

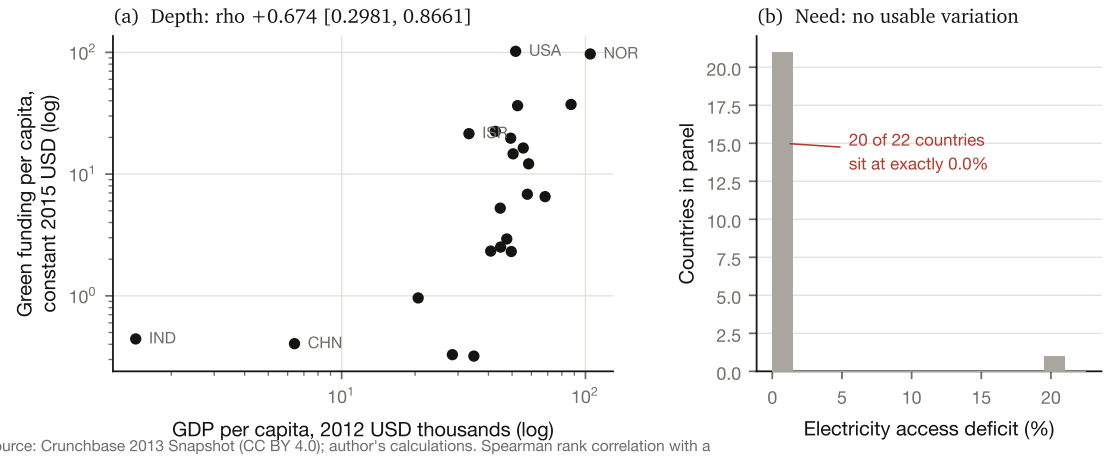
Green funding is extraordinarily concentrated: a Gini of 0.917, with the top five countries holding 89.9% of all green dollars across 42 countries.

The need limb cannot be tested. Twenty of the 22 panel countries record an electricity-access deficit of exactly zero; the regressor is in substance a dummy for one country. An income-controlled coefficient on it appeared positive and significant, met the pre-registered falsification condition, and survived both leave-one-out and collinearity checks before a degenerate-variance check rejected it.

Verdict: Partially supported.

Capital tracks market depth; whether it tracks need cannot be measured here

Green funding intensity across 22 countries with at least three green deals, 2000–2012.



Source: Crunchbase 2013 Snapshot (CC BY 4.0); author's calculations. Spearman rank correlation with a bootstrap interval over 10,000 draws; n = 22 countries. The planned small-multiple of need indicators is not drawn: panel (b) shows why. A scatter against a regressor on which 20 of 22 countries record exactly zero would read as a relationship and is not one. Green funding is highly concentrated, with a Gini of 0.917 and the top five countries holding 89.9% of green dollars.

Figure 9. Green funding intensity against market depth, beside the need regressor's distribution.

What this shows about the record: countries with real energy-access deficits have too few green deals to enter the panel. The need indicators were not tested and found null; they could not be tested.

6.6 Six routes, one conclusion

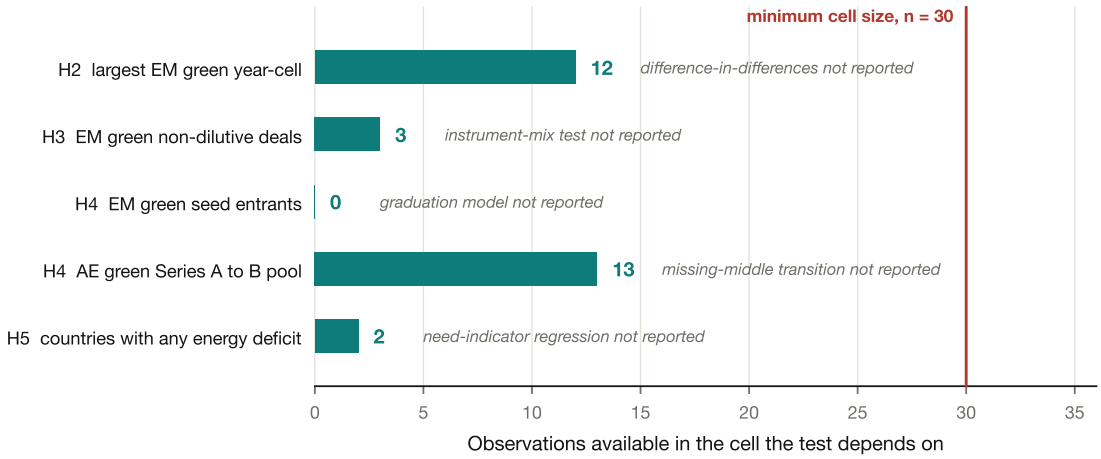
Three hypotheses were unresolved, each for a different stated reason: H2′s identifying assumption failed, H4 had 13 companies at the transition it exists to measure, H5′s need regressor had no variance. Three further failures are properties of the record rather than of any hypothesis: 88.2% of emerging-market green deals carry no specific sub-domain; a pre-declared venture-stage criterion failed because project finance is coded as venture rounds; and H3′s emerging-market arm is biased by that same coding in a direction stated in advance.

Figure 10 puts the first four on one scale, against the minimum cell size fixed in config before any hypothesis was tested. No cell clears it.

These are six independent routes to one conclusion. The open venture record cannot support the capital-regime analysis the field needs.

Every emerging-market test fails on the same wall, for different reasons

Observations available against the pre-registered minimum cell size, across H2–H5.



Source: Crunchbase 2013 Snapshot (CC BY 4.0); author's calculations. The minimum cell size of 30 was fixed in config before any hypothesis was tested, and no bar clears it. These are five distinct failures rather than one repeated: a difference-in-differences with no usable time series; an instrument-mix comparison resting on 3 non-dilutive deals, drawn from an emerging-market green sample of 34 which does itself clear the threshold; a graduation model with no entrants at all; a missing-middle transition holding 13 companies even in the best-covered bloc; and a need regressor on which 20 of 22 countries record exactly zero. Three

Figure 10. Observations available in each emerging-market test cell, against the minimum.